
EE1060: Differential Equations and Transform Techniques

Lecture 18: Boundary Value Problems

Topics :

1. Dirichlet Boundary Condition
 2. Periodic Boundary Condition
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Boundary Value Problems

A 2nd order ODE

$$y'' + p(x)y' + q(x)y = g(x) \quad \text{with}$$

$$y(0) = y_0 \quad \text{and}$$

$$y(L) = y_L$$

typical Boundary Value Problem

Conditions are specified at multiple points in the domain.

$$\{y, y', \dots, y^{(n-1)}\}$$

Example:-

$$\frac{d^2 y}{dx^2} + \lambda y = 0 \rightarrow \textcircled{1}$$

with Boundary Cond specified at 0, L.

→ $\textcircled{1}$ Dirichlet Boundary Cond : $y(0), y(L)$

Trivial solution $y \equiv 0$

Try to see if $\exists \lambda$, for which $\textcircled{1}$ has non-trivial sol

$\lambda \leftarrow$ eigen values.

$y(x) \leftarrow$ eigen function

$\textcircled{2}$ Neumann " " : $y'(0), y'(L)$

$\textcircled{3}$ Mixed : $y'(0), y(L)$
or $y(0), y'(L)$

→ $\textcircled{4}$ Periodic : $y(0) = y(L)$
 $y'(0) = y'(L)$

Dirichlet Boundary Condition

$$\frac{d^2 y}{dx^2} + \lambda y = 0 \quad \text{with BC} \quad y(0) = 0 \quad \text{and} \quad y(L) = 0.$$

Case I:- $\lambda = 0$, $\frac{d^2 y}{dx^2} = 0 \Rightarrow y(x) = C_1 + C_2 x$.

$$\left. \begin{array}{l} y(0) = 0 \Rightarrow C_1 = 0 \\ y(L) = 0 \Rightarrow C_2 = 0 \end{array} \right\} y \equiv 0 \text{ (Trivial Sol.)}$$

Case II:- $\lambda = -\omega_n^2$; auxiliary eqn: $s^2 - \omega_n^2 = 0 \Rightarrow s = \pm \omega_n$

$$y(x) = C_1 e^{\omega_n x} + C_2 e^{-\omega_n x}.$$

$$y(0) = 0 \Rightarrow C_1 + C_2 = 0 \Rightarrow C_2 = -C_1$$

$$y(x) = C_1 (e^{\omega_n x} - e^{-\omega_n x}) = 2C_1 \sinh(\omega_n x).$$

$$y(L) = 0 \Rightarrow 2C_1 \sinh(\omega_n L) = 0 \Rightarrow \underbrace{C_1 = 0}_{y \equiv 0} \quad (\text{or } \omega_n = 0).$$

Dirichlet Boundary Condition

Case III :- $\lambda = \omega_n^2$ $S = f \circ \omega_n$.

$$y(x) = C_1 \cos \omega_n x + C_2 \sin \omega_n x$$

$$y(0) = 0 \Rightarrow C_1 = 0 \Rightarrow y(x) = C \sin \omega_n x.$$

$$y(L) = 0 \Rightarrow C \sin(\omega_n L) = 0. \longrightarrow C = 0 \rightarrow \text{Trivial sol}$$

$$\omega_n L = 2\pi n \rightarrow \text{non trivial sol.}$$

$$\omega_n = \frac{\pi n}{L}$$

$$y(x) = C \sin\left(\frac{\pi n}{L} x\right) \text{ for } \lambda = \omega_n^2 = \left(\frac{\pi n}{L}\right)^2 \text{ when } n = 1, 2, \dots$$

Periodic Boundary Condition

$$\frac{d^2 y}{dx^2} + \lambda y = 0 \quad \text{with BC} \quad y(0) = y(L) \quad \text{and} \quad y'(0) = y'(L)$$

Case I:- $\lambda = 0 \Rightarrow \frac{d^2 y}{dx^2} = 0 \Rightarrow y(x) = c_1 + c_2 x.$

$$\left. \begin{aligned} y(0) = y(L) &\Rightarrow c_1 = c_1 + c_2 L \Rightarrow c_2 = 0. \\ y(x) = c &\text{ \& it satisfies } y'(0) = y'(L) \end{aligned} \right\} \begin{aligned} \lambda = 0 \text{ \& } \\ y(x) = c. \end{aligned}$$

Case II:- $\lambda = -\omega_n^2 \quad y(x) = c_1 e^{\omega_n x} + c_2 e^{-\omega_n x}$

$$\text{BC: } y(0) = y(L) \Rightarrow c_1 + c_2 = c_1 e^{\omega_n L} + c_2 e^{-\omega_n L} \longrightarrow \textcircled{1}$$

$$y'(0) = y'(L) \Rightarrow \omega_n c_1 - \omega_n c_2 = \omega_n c_1 e^{\omega_n L} - \omega_n c_2 e^{-\omega_n L}.$$

$$\Rightarrow c_1 - c_2 = c_1 e^{\omega_n L} - c_2 e^{-\omega_n L} \longrightarrow \textcircled{2}$$

Sol: $\omega_n L = 0 \Rightarrow$ not of interest
No non-trivial sol exists.

Periodic Boundary Condition

Case III :- $\lambda = \omega_n^2$ $y(x) = C_1 \cos(\omega_n x) + C_2 \sin(\omega_n x) \rightarrow$

$$y(0) = y(L) \Rightarrow C_1 = C_1 \cos(\omega_n L) + C_2 \sin(\omega_n L) \rightarrow \textcircled{1}$$

$\textcircled{1}$ holds when $\omega_n L = 2\pi n \Rightarrow y(x) = C_2 \sin\left(\underbrace{\frac{2\pi n}{L} x}_{\omega_n}\right) + C_1 \cos\left(\underbrace{\frac{2\pi n}{L} x}_{\omega_n}\right)$

Be $\textcircled{1}$: $y'(0) = y'(L) \Rightarrow C_2 \omega_n = C_2 \omega_n \cos(\omega_n L) - C_1 \omega_n \sin(\omega_n L) \rightarrow \textcircled{2}$

$\textcircled{2}$ holds when $\omega_n = \frac{2\pi n}{L}$

eigen value = $\lambda = \omega_n^2 = \left(\frac{2\pi n}{L}\right)^2$

eigen function = $\begin{cases} \sin(\omega_n x) = \sin\left(\frac{2\pi n}{L} x\right) \\ \cos(\omega_n x) = \cos\left(\frac{2\pi n}{L} x\right) \end{cases}$